MCS-211	ADVANCED SOFTWARE ENGINEERING	L T P 3 1 0	4 Credits

Course Objectives:

CO1	An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
CO2	An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.
CO3	an ability to use the techniques, skills, and modern engineering tools and processes necessary for software engineering practice.
CO4	An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
CO5	An ability to communicate effectively with a range of audiences

MODULE I

Software project management, metric and management, software configuration management, software risk management, requirements engineering,

MODULE II

Software quality assurance, software reliability models, object oriented design, Unified Modeling Language, (UML), Use case modeling

MODULE III

Jakson method for design, case tools and technology, clean room method for software development, real time software specification and design.

References

1. *Sommerville, "Software Engg. Principles and Practices," Addison-Wesley.*
2. *Roger S Pressman, "Software Engg," McGraw-Hill.*
3. *Pankaj Jalote, "Introduction to Software Engineering," Springer.*

Course Outcome:

CO1	Understand and adhere to professional ethical standards in the system development and modification process, especially by accepting responsibility for the consequences of design decisions and design implementations
CO2	The ability to build and configure major operating system components
CO3	The ability to analyze and implement solutions to complex problems involving computers and networks
CO4	The ability to work effectively in teams
CO5	A solid understanding to the methods of modern software engineering